🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.11

Graph Inequalities

Lesson 7-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes **Level 2 Standard**

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the solutions of an inequality on a number line.
Language Objective: I can explain my graph using the words number line, open circle, closed circle, and solution set.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Number line | **Open circle** | **Closed circle** | **Solution set** | **Inequality** | **Boundary point**

👆 *Tap any word to see what it means and a picture.*

1. A line with numbers in order used to show values and solutions is a ______________.

2. A hollow circle that shows a value is NOT included, used for < or >, is an ______________.

3. A filled-in circle that shows a value IS included, used for ≤ or ≥, is a ______________.

4. All the values that make an inequality true form the ______________.

5. A math sentence comparing two amounts with <, >, ≤, or ≥ is an ______________.

6 The number where the solution set begins on the number line is the

______________________ .

Reflect — Exit Ticket

How would you graph the inequality $x \geq 9$?

A. Closed circle at 9, shaded right

B. Open circle at 9, shaded right

C. Closed circle at 9, shaded left

D. Open circle at 9, shaded left

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Open circle
3. **Notes 3:** Closed circle
4. **Notes 4:** Solution set
5. **Notes 5:** Inequality
6. **Notes 6:** Boundary point
7. **Watch for this mistake:** *A common mistake in Graph Inequalities is skipping the key idea: "Closed circle for ≤ or ≥ (included), open circle for < or > (not included); shade right for greater, left for less." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Open circle at 15, shade right — *'Greater than' (>) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.*
9. **Try It 2:** A. Closed circle at 4, shade left — *'Less than or equal to' (≤) includes 4 — closed circle. Smaller values work, so shade left.*
10. **Exit Ticket:** A. Closed circle at 9, shaded right — *$x \geq 9$ means 9 is included (closed circle) and values are 9 or greater (shaded right).*